The documentation and process conversion measures necessary to comply with this document shall be completed by 16 November 2023.

INCH-POUND

MIL-PRF-19500/355W
w/AMENDMENT 1
16 August 2023
SUPERSEDING
MIL-PRF-19500/355W
14 February 2020

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, UNITIZED DUAL, NPN, SILICON, TYPES 2N2919, 2N2920, JAN, JANTX, JANTXV, JANS, JANHC AND JANKC.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for two electrically isolated, matched NPN radiation hardened silicon transistors as one dual unit. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device type, and two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device, as specified in MIL-PRF-19500. Provisions for radiation hardness assurance (RHA) to eleven radiation levels ("E", "K", "U", "M", "D", "P", "L", "R", "F", "G" and "H") are provided for JANTXV, JANS, JANHC, JANKC, JANHCD, and JANKCD product assurance levels.

1.2 Package outlines and die topography. The device package for the encapsulated device type are as follows: similar to TO-78 in accordance with figure 1, surface mount version in accordance with figure 2. The dimensions and topography for JANHC and JANKC unencapsulated die are as follows: A version die in accordance with figure 3, B version die in accordance with figure 4, and C and D version dies in accordance with figure 5.

1.3 Maximum ratings. Unless otherwise specified, T_C =+25°C.

Type	I_C	V_{CBO}	V_{CEO}	V_{EBO}	T_J and T_{STG}
	mA dc	V dc	V dc	V dc	°C
All types	30	70	60	6	-65 to +200

(1) For T_A > +25°C, derate linearly 1.143 mW/°C, one section; 2.000 mW/°C, both sections.
(2) For T_C > +25°C, derate linearly 1.714 mW/°C, one section; 2.571 mW/°C, both sections.

P_T (1) T_A = +25°C		P_T (2) T_C = +25°C		$R_{\theta JA}$		$R_{\theta JC}$	
One section	Both sections	One section	Both sections	One section	Both sections	One section	Both sections
mW	mW	mW	mW	°C/W	°C/W	°C/W	°C/W
200	350	300	450	875	500	583	388

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1.4 Primary electrical characteristics of each individual section. Unless otherwise specified, T_C =+25°C.

	h_{FE1}	h_{FE1}	$\|h_{fe}\|$	$V_{CE(SAT)}$
	V_{CE} = 5 V dc I_C = 10 µA dc	V_{CE} = 5 V dc I_C = 10 µA dc	V_{CE} = 5 V dc I_C = 0.5 mA dc f = 20 MHz	I_C = 1 mA dc I_B = 100 µA dc
	2N2919 2N2919L 2N2919U	2N2920 2N2920L 2N2920U		
				V dc
Min	60	175	3.0	
Max	240	600	20	0.3

1.5 Primary electrical matching characteristics of each individual section. Unless otherwise specified, T_C =+25°C.

	$\frac{h_{FE2-1}}{h_{FE2-2}}$ V_{CE} = 5 V dc I_C = 100 µA dc	$\|V_{BE1} - V_{BE2}\|_1$ V_{CE} = 5 V dc I_C = 10 µA dc	$\|\Delta(V_{BE1} - V_{BE2})_{\Delta TA}\|_1$ V_{CE} = 5 V dc I_C = 100 µA dc T_A = +25°C and -55°C	$\|\Delta(V_{BE1} - V_{BE2})_{\Delta TA}\|_2$ V_{CE} = 5 V dc I_C = 100 µA dc T_A = +125°C and +25°C
		mV dc	mV dc	mV dc
Min	0.9			
Max	1.1	5	0.8	1.0

1.6 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.6.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, “JANTXV” and "JANS".

1.6.2 JAN certification mark and quality level for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

* 1.6.3 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest for JANTXV and JANS quality levels are as follows: “E”, “K”, “U”, “M”, “D”, “P“, “L”, “R”, “F”, “G” and “H”.

1.6.4 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.6.4.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

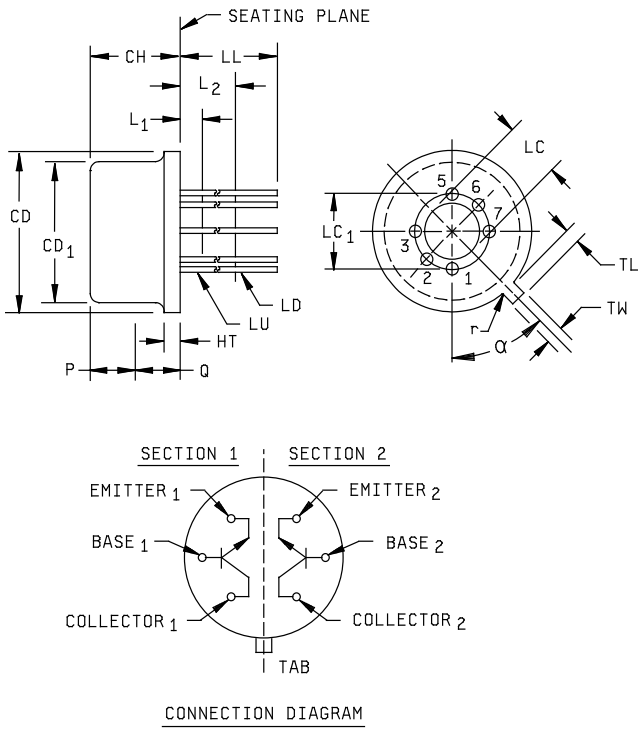
1.6.4.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "2919" and "2920".

1.6.4.3 Suffix symbols. The suffix letter "L" is used on devices that are packaged in the TO-78 package of figure 1 that have a long lead length: 1.500 inches (38.10 mm) minimum and 1.750 inches (44.45 mm) maximum. Devices with no suffix have standard length leads of 0.5 inch (12.7 mm) minimum to .75 inch (19.1 mm) maximum in figure 1. The suffix letter “U” is used on devices that are packaged in the surface mount package of figure 2.

1.6.5 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A" "B", “C”, and “D” (see figure 3, figure 4, figure 5 and 6.5).

1.6.6 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

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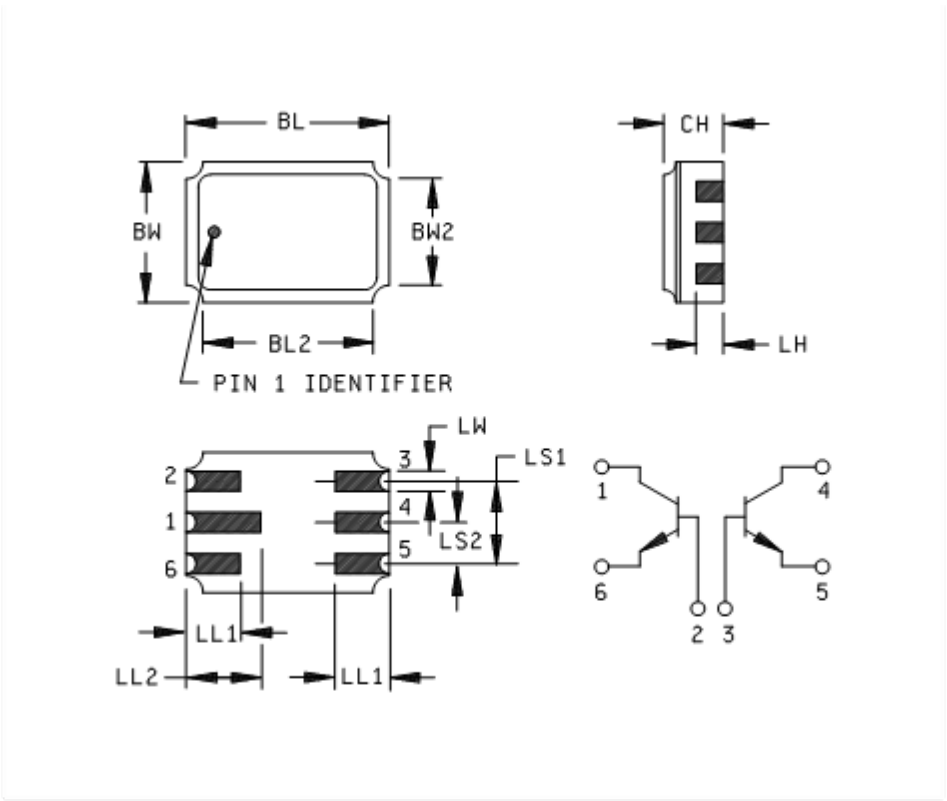
CONNECTION DIAGRAM

	Dimensions				
Symbol	in	in	mm	mm	Notes
	Min	Max	Min	Max	
CD	.335	.370	8.51	9.40	
CD_1	.305	.335	7.75	8.51	
CH	.140	.260	3.56	6.60	
HT	.009	.041	0.23	1.04	
LC	.140	.160	3.56	4.06	
LC_1	.200	TP	5.08	TP	9
* LD	.016	.021	0.41	0.53	10
LL	See notes 10, 11, and 12				
LU	.016	.019	0.41	0.48	10
L_1		.050		1.27	10
L_2	.250		6.35		10
P	.100		2.54		8
Q		.050		1.27	7
TL	.029	.045	0.74	1.14	5, 6
TW	.028	.034	0.71	0.86	4, 5
r		.010		0.25	
α	45°	TP	45°	TP	9

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Tab shown omitted.
4. Lead numbers 4 and 8 are omitted on this variation.
5. Beyond r maximum, TW shall be held to a minimum length of .21 inch (5.33 mm).
6. TL shall be measured from maximum CD.
7. Details of outline in this zone are optional.
8. CD_1 shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
9. Leads at gauge plane .054 - .055 inch (1.37 - 1.40 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at a maximum material condition (MMC) relative to the tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedures described on gauge drawing GS-1.
10. LU applies between L_1 and L_2. LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
11. For transistor types 2N2919 and 2N2920, LL is .500 inch (12.70 mm) minimum and .750 inch (19.05 mm) maximum.
12. For transistor type 2N2919L and 2N2920L, LL is 1.500 inches (38.10 mm) minimum and 1.750 inches (44.45 mm) maximum.

FIGURE 1. Physical dimensions 2N2919, 2N2919L, 2N2920, and 2N2920L (TO-78).

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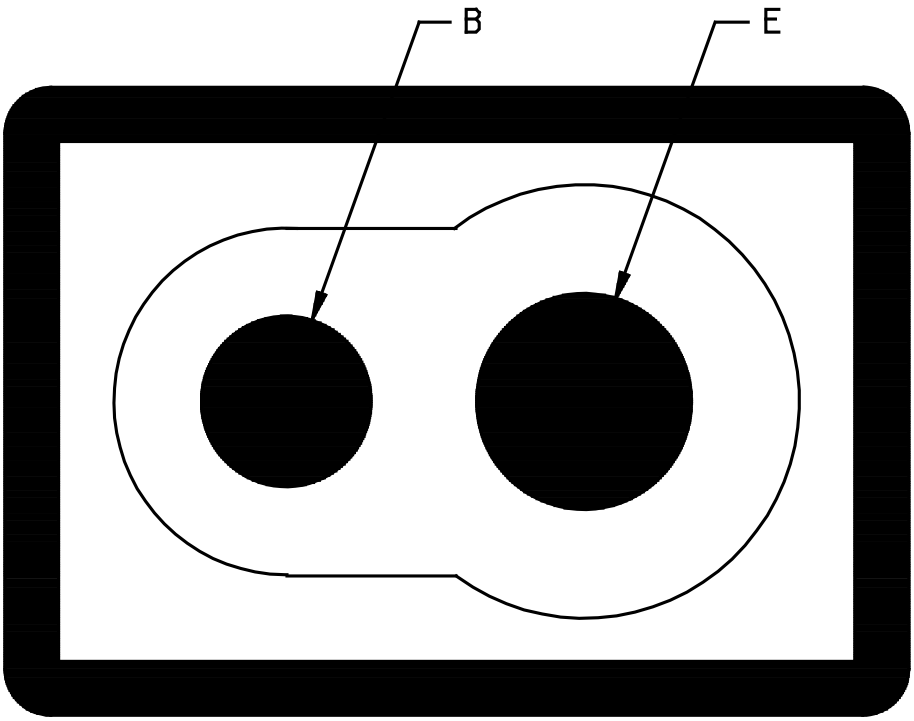
Symbol	Dimensions			
	in	in	mm	mm
	Min	Max	Min	Max
BL	.240	.250	6.10	6.35
BL_2		.250		6.35
BW	.165	.175	4.19	4.44
BW_2		.175		4.44
CH	.044	.080	1.12	2.03
LH	.026	.039	0.66	0.99
LL_1	.060	.070	1.52	1.78
LL_2	.082	.098	2.08	2.49
LS_1	.095	.105	2.41	2.67
LS_2	.045	.055	1.14	1.39
LW	.022	.028	0.56	0.71

Pin no.	Transistor
1	Collector no. 1
2	Base no. 1
3	Base no. 2
4	Collector no. 2
5	Emitter no. 2
6	Emitter no. 1

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.

FIGURE 2. Physical dimensions (2N2919U and 2N2920U) surface mount.

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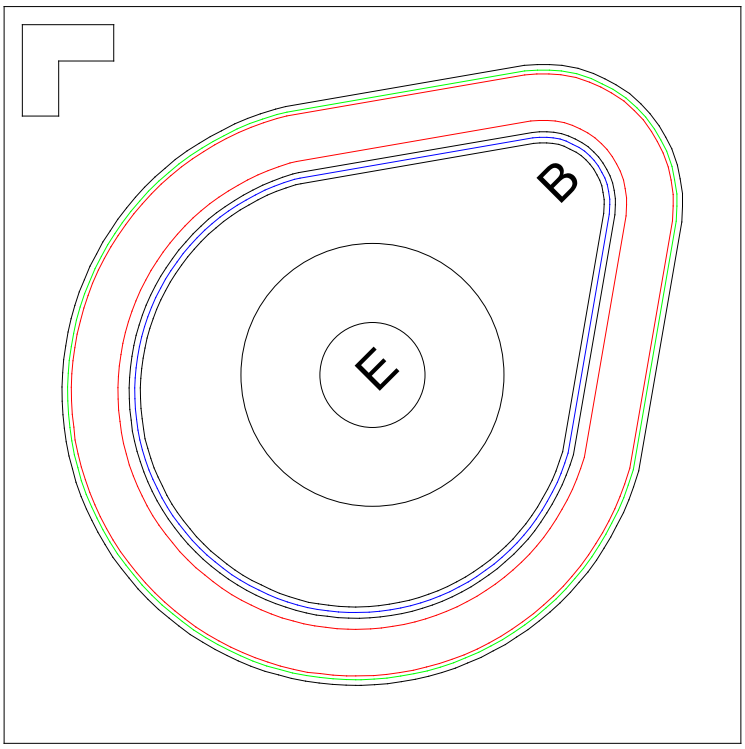


NOTES:

1. Chip size ..... .015 x .019 inch ±.001 inch (0.381 x 0.4826 mm ±0.0254 mm).
2. Chip thickness ..... .010 ±.0015 inch (0.254 ±0.0381 mm).
3. Top metal ..... Aluminum 15,000Å minimum, 18,000Å nominal.
4. Back metal ..... A. Gold 2,500Å minimum, 3,000Å nominal.
   B. Eutectic Mount - No Gold.
5. Backside ..... Collector.
6. Bonding pad ..... B = .003 inch (0.0762 mm), E = .004 inch (0.1016 mm) diameter.
7. Passivation ..... Si_3N_4 (Silicon Nitride) 2 kÅ min, 2.2 kÅ nom.

FIGURE 3. Physical dimensions (JANHCA and JANKCA die).

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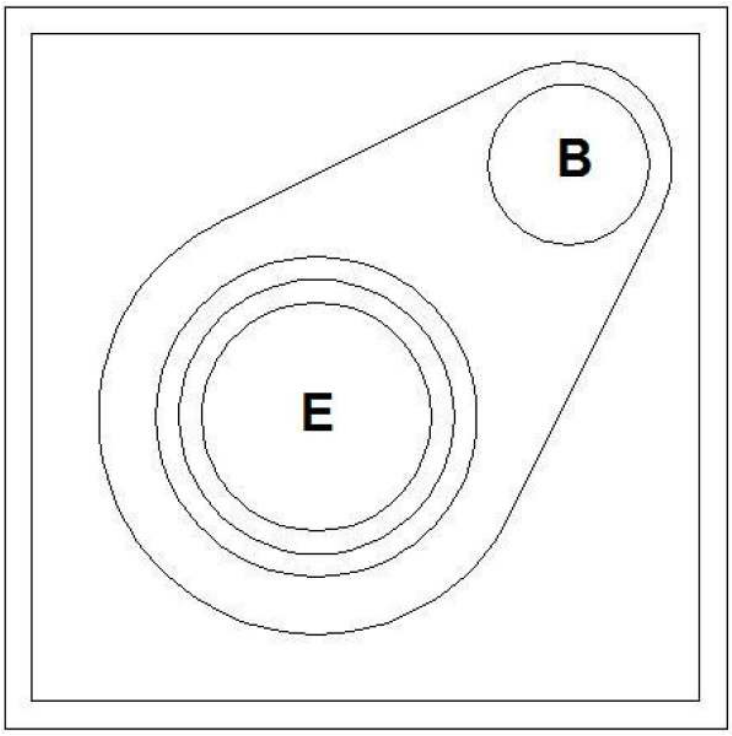


NOTES:

1. Die size--------- .018 x .018 inch (0.457 mm x 0.457 mm).
2. Die thickness--- .008 ±.0016 inch (0.203 mm ±0.04 mm).
3. Base pad-------- .0025 inch diameter (0.06 mm).
4. Emitter pad----- .003 inch diameter (0.076 mm).
5. Back metal----- Gold, 6,500 ±1,950Å.
6. Top metal------ Aluminum, 19,500 ±2,500Å.
7. Back side------ Collector.
8. Glassivation--- SiO_2, 7,500 ±1,500Å.

FIGURE 4. Physical dimensions (JANHCB and JANKCB) B version die.

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Die Size:	.023 x .023 Inch ± .002 Inch (0.584 mm x 0.584 mm ± 0.0508)
Die Thickness:	.010 Inch ± .002 Inch (0.254 mm ± 0.0508 mm)
Top Metal:	Aluminum, 16,000 Å Minimum, 24,000 Å Nominal
Back Metal:	Gold, 4,500 Å Minimum, 5,500 Å Nominal
Backside:	Collector
Bonding Pads:	Base = .003 inch (0.08mm)
	Emitter = .004 (0.10 mm)

FIGURE 5. Physical dimensions (JANHCC, JANKCC, JANHCD, and JANKCD) C and D version dies.

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* 1.7 Radiation features. The following radiation features are applicable for RHA devices supplied to this specification sheet.

* 1.7.1 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-PRF-750 are as follows:

For device type 2N2919, L, U, 2N2920, L, U: ..... 300 krads(Si) 1/

* 1.7.2 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition D (dose rate ≤ 10 mrad(Si)/s) of method 1019 of MIL-PRF-750 are as follows:

For device type 2N2919, L, U, 2N2920, L, U: ..... 100 krads(Si) 2/

* 1/ The manufacturers supplying these device types have performed characterization testing in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-PRF-750. The radiation end point limits are guaranteed only to a maximum TID level of 300 krads(Si).
* 2/ The manufacturers supplying these device types have performed characterization testing in accordance with condition D (dose rate ≤ 10 mrad(Si)/s) of method 1019 of MIL-PRF-750. The radiation end point limits are guaranteed only to a maximum TID level of 100 krads(Si).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

h_{FE-1}/h_{FE-2}	Static forward-current-gain-ratio. The matching ratio of the static forward-current transfer ratio of each section.
$R_{\theta JA}$	Thermal resistance junction to ambient.
$R_{\theta JSP(IS)}$	Thermal resistance junction to solder pads (infinite sink mount to PCB).
$\lvert V_{BE1} - V_{BE2} \rvert$	Absolute value of base-emitter-voltage differential between the individual transistors (or sections, see figure 1 and figure 2).
$\lvert \Delta V_{BE1-2}$ (T1) - ΔV_{BE1-2} (T2)$\rvert$	Absolute value of the algebraic difference between the base-emitter-voltage differentials between the individual sections at two different temperatures.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1, 2, 3, 4 and 5. No lead (Pb) shall be used in the construction of the die bonds.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements, PIN designators, and test levels shall be as defined in MIL-PRF-19500 and 6.5.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, 1.5, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

* 3.8 Marking. Marking shall be in accordance with MIL-PRF-19500. The radiation hardened designator E, K, U, M, D, P, L, R, F, G, or H shall immediately precede (or replace) the device “2N” identifier (depending upon degree of abbreviation required).

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I, table II, and table III).

d. Element evaluation (see 4.5.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC, JANKC, JANHCD, and JANKCD die. Qualification shall be in accordance with MIL-PRF-19500.

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4.3 Screening of encapsulated devices (JANS, JANTXV, and JANTX levels only). Screening of packaged devices shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANS level	Measurements for JANTX and JANTXV levels
(1) 3c	Thermal impedance, method 3131 of MIL-STD-750	Thermal impedance, method 3131 of MIL-STD-750
9	I_{CBO2}, h_{FE3}, $\frac{h_{FE2-1}}{h_{FE2-2}}$	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CBO2}, h_{FE3}, $\frac{h_{FE2-1}}{h_{FE2-2}}$ ΔI_{CBO2} = 100 percent of initial value or 1 nA dc, whichever is greater. Δh_{FE3} = ±20 percent	I_{CBO2}, h_{FE3}, $\frac{h_{FE2-1}}{h_{FE2-2}}$
12	See 4.3.2	See 4.3.2
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 1 nA dc, whichever is greater; Δh_{FE3} = ±25 percent	Subgroup 2 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 1 nA dc, whichever is greater; Δh_{FE3} = ±25 percent

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements for JANTX and JANTXV levels.

4.3.1 Thermal impedance. The thermal impedance shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining V_H, V_{CE}, I_M, I_H, t_H, and t_{MD}. The limit used in screen 3c and table I, subgroup 2 and figure 6, and figure 7 shall be set statistically by the supplier over several die lots and submitted to the qualifying activity for approval.

4.3.2 Power burn-in conditions. V_{CB} = 10 - 30 V dc. Power shall be applied to achieve P_D = 100 percent of P_T maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table E-VIB, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) of MIL-PRF-19500 and 4.4.2.1 herein. Delta requirements shall be in accordance with and 4.5.9 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with 4.5.9 herein.

4.4.2.1 Group B inspection, table E-VIA (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 - 30 V dc.
B5	1027	V_{CB} = 10 - 30 V dc, $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours min, sample size in accordance with table E-VIA of MIL-PRF-19500, adjust T_A to achieve T_J = +275°C minimum. Option 2: 216 hours min., sample size = 45, c = 0; adjust T_A to achieve T_J = +225°C minimum.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failures mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 - 30 V dc, power shall be applied and ambient temperature T_A adjusted to achieve T_J = +150°C minimum using a minimum of P_D = 100 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall be in accordance with all of the following requirements:

a. Samples shall be selected from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection.

b. When the final lead finish is solder, or any plating prone to oxidation at high temperature, the samples for life test (Step 1) may be tested prior to the application of final lead finish.

c. Separate samples may be used for each step.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Delta requirements shall be in accordance with 4.5.9 herein.

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4.4.3.1 Group C inspection, table E-VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, not applicable to surface mount.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and 4.3.1.
C6	1026	1,000 hours at V_{CB} = 10 - 30 V dc, power shall be applied and ambient temperature T_A adjusted to achieve T_J = +150°C minimum using a minimum of P_D = 100 percent of maximum rated P_T as defined in 1.3. n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

4.4.3.2 Group C inspection, table E-VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, not applicable to surface mount.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and 4.3.1.
C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750 for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence, as applicable (see 6.2 herein) except group D, subgroup 2 may be performed separate from other subgroups. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

* 4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Delta measurements shall be in accordance with the applicable steps of 4.5.9.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Testing of units. All specified electrical tests, including end-point tests, shall be performed equally on both sections of the transistor types covered herein, except where the electrical characteristic being evaluated applies to the transistor as a device entity.

4.5.3 Element evaluation of unencapsulated die. The element evaluation of unencapsulated die shall be in accordance with appendix G of **Error! Reference source not found.**. For subgroup 4, the burn-in duration for the JANKC level shall follow the JANS requirements and the JANHC shall follow the JANTX requirements.

4.5.4 Disposition of leads when testing characteristics of each section. During the measurement of the characteristics of each section, the leads of the section not under test shall be open-circuited.

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4.5.5 Forward-current-gain ratio. The value for the forward-current-gain ratio for each individual section of a dual unit shall be measured using method 3076 of MIL-STD-750. The forward-current-gain ratio shall be calculated by dividing one of the values by the other. If possible, this ratio shall be measured directly to improve accuracy.

4.5.6 Base-emitter-voltage differential. The base-emitter-voltage differential shall be determined by connecting the emitters of the individual sections together, applying specified electrical test conditions to each individual section in accordance with method 3066 of MIL-STD-750, test condition B, and measuring the absolute value of the voltage between the bases of the individual sections of a dual unit.

4.5.7 Base-emitter-voltage differential change with temperature. The value of the base-emitter-voltage differential shall be measured at the two specified temperatures in accordance with 4.5.6 except that the polarities of the differentials and identities of the individual sections shall be maintained. The absolute value of the algebraic difference between the values at the two temperature extremes shall be calculated. A mathematical formula for this parameter is:

$$|(V_{BE1}(T_1) - V_{BE2}(T_1)) - (V_{BE1}(T_2) - V_{BE2}(T_2))|$$

4.5.8 Noise figure test. Noise figure shall be measured using Quan Tech Laboratories test set model no. 2173C/2181, or equivalent. Conditions shall be as specified in table I.

4.5.9 Delta requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit	Limit
		Method	Conditions		Min	Max
1	Collector-base cutoff current	3036	Bias condition D, V_{CB} = 45 V dc	ΔI_{CBO2}	100 percent of initial value or 1 nA dc, whichever is greater.	
2	Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 mA dc; pulsed, see 4.5.1	Δh_{FE3}	±25 percent change from initial reading.	

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071					
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Salt atmosphere (corrosion, laser marked devices only) 4/	1041	n = 6 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal 4/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hours or T_A = +300°C at t = 2 hours n = 11 wires, c = 0				
Decap internal visual (design verification) 4/	2075	n = 4 devices, c = 0				
Subgroup 2						
Thermal impedance 6/	3131	See 4.3.1	$Z_{\theta JX}$			°C/W
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 70 V dc	I_{CBO1}		10	µA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 6 V dc	I_{EBO1}		10	µA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	60		V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 45 V dc	I_{CBO2}		2	nA dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 5 V dc	I_{CEO1}		2	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO2}		2	nA dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 µA dc	h_{FE1}			
2N2919, 2N2919L, 2N2919U				60	240	
2N2920, 2N2920L, 2N2920U				175	600	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 100 µA dc	h_{FE2}			
2N2919, 2N2919L, 2N2919U				100	325	
2N2920, 2N2920L, 2N2920U				235	800	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 mA dc;	h_{FE3}			
2N2919, 2N2919L, 2N2919U				150	600	
2N2920, 2N2920L, 2N2020U				300	1,000	
Base-emitter saturation voltage	3066	Test condition A; I_C = 1.0 mA dc; I_B = 100 µA dc;	$V_{BE(sat)1}$	0.5	1.0	V dc
Collector-emitter saturation voltage	3071	I_C = 1.0 mA dc; I_B = 100 µA dc;	$V_{CE(sat)1}$		0.3	V dc
Forward-current transfer ratio (gain ratio)	3076	V_{CE} = 5 V dc, I_C = 100 µA dc (see 4.5.5)	$\frac{h_{FE2-1}}{h_{FE2-2}}$	0.9	1.1	

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Absolute value of base-emitter-voltage differential	3066	Test condition B; V_{CE} = 5 V dc, I_C = 10 μA dc (see 4.5.6)	$\lvert V_{BE1} - V_{BE2} \rvert_1$		5	mV dc
Absolute value of base-emitter-voltage differential	3066	Test condition B; V_{CE} = 5 V dc, I_C = 100 μA dc (see 4.5.6)	$\lvert V_{BE1} - V_{BE2} \rvert_2$		3	mV dc
Absolute value of base-emitter-voltage differential	3066	Test condition B; V_{CE} = 5 V dc, I_C = 1 mA dc (see 4.5.6)	$\lvert V_{BE1} - V_{BE2} \rvert_3$		5	mV dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Base-emitter-voltage (nonsaturated) (absolute value of differential change with temperature)	3066	Test condition B; V_{CE} = 5 V dc, I_C = 100 μA dc T_A = +125°C and +25°C (see 4.5.7)	$\lvert \Delta V_{BE1} - V_{BE2} \Delta T_A \rvert_2$		1	mV dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 45 V dc	I_{CBO3}		2.5	μA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio 2N2919, 2N2919L, 2N2919U 2N2920, 2N2920L, 2N2920U	3076	V_{CE} = 5 V dc; I_C = 10 μA dc	h_{FE4}	 20 50		
Base-emitter-voltage (nonsaturated) (absolute value of differential change with temperature)	3066	Test condition B; V_{CE} = 5 V dc, I_C = 100 μA dc T_A = +25°C and -55°C (see 4.5.7)	$\lvert \Delta V_{BE1} - V_{BE2} \Delta T_A \rvert_1$		0.8	mV dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Small-signal short-circuit input impedance	3201	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 1 kHz	h_{ie}	3	30	kΩ
Small-signal open-circuit reverse voltage transfer ratio	3211	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 1 kHz	h_{re}		1×10^{-3}	
Small-signal open-circuit output admittance	3216	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 1 kHz	h_{oe}		60	μmhos
Small-signal short-circuit forward current transfer ratio (magnitude h_{fe})	3306	V_{CE} = 5 V dc; I_C = 0.5 mA dc; f = 20 MHz	$\lvert h_{fe} \rvert$	3	20	
Open circuit output capacitance	3236	V_{CB} = 5 V dc; I_E = 0 100 kHz ≤ f ≤ 1 MHz	C_{obo}		5	pF
Noise figure	3246	V_{CE} = 5 V dc, I_C = 10 μA dc R_g = 10 kΩ, (see 4.5.8)				
Test 1 Test 2 Test 3		f = 100 Hz f = 1 kHz f = 10 kHz	F1 F2 F3		5 3 3	dB dB dB
Subgroup 5						
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 40 V dc	I_{CES}		20	nA dc
Subgroups 6 and 7						
Not required						

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, steps 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroups 1.

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* TABLE II. Group D inspection.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 1 4/						
Neutron irradiation	1017	Neutron exposure V_{CES} = 0 V				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 70 V dc	I_{CBO1}		20	µA dc
Cutoff current, emitter to base	3061	Bias condition D; V_{EB} = 6 V dc	I_{EBO1}		20	µA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	60		V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 45 V dc	I_{CBO2}		4	nA dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 5 V dc	I_{CEO1}		4	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO2}		4	nA dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 µA dc	[h_{FE1}] 5/			
2N2919, 2N2919L, 2N2919U				[30]	240	
2N2920, 2N2920L, 2N2920U				[87.5]	600	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 100 µA dc	[h_{FE2}] 5/			
2N2919, 2N2919L, 2N2919U				[50]	325	
2N2920, 2N2920L, 2N2920U				[117.5]	800	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 mA dc;	[h_{FE3}] 5/			
2N2919, 2N2919L, 2N2919U				[75]	600	
2N2920, 2N2920L, 2N2920U				[150]	1,000	

See footnotes at end of table.

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* TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 1 4/ - Continued.						
Absolute value of base-emitter-voltage differential	3066	Test condition B; V_{CE} = 5 V dc, I_C = 1 mA dc (see 4.5.6)	$\lvert V_{BE1} - V_{BE2} \rvert_3$		10	mV dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 1.0 mA dc; I_B = 100 µA dc;	$V_{BE(sat)1}$	0.50	1.15	V dc
Collector-emitter saturation voltage	3071	I_C = 1.0 mA dc; I_B = 100 µA dc;	$V_{CE(sat)1}$		0.35	V dc
Forward-current transfer ratio (gain ratio)	3076	V_{CE} = 5 V dc, I_C = 100 µA dc (see 4.5.6)	$\frac{h_{FE2-1}}{h_{FE2-2}}$	0.8	1.25	
Absolute value of base-emitter-voltage differential	3066	Test condition B; V_{CE} = 5 V dc, I_C = 10 µA dc (see 4.5.6)	$\lvert V_{BE1} - V_{BE2} \rvert_1$		10	mV dc
Absolute value of base-emitter-voltage differential	3066	Test condition B; V_{CE} = 5 V dc, I_C = 100 µA dc (see 4.5.6)	$\lvert V_{BE1} - V_{BE2} \rvert_2$		6	mV dc
Subgroup 2						
* Total dose irradiation	1019	Gamma exposure V_{CES} = 40 V Condition A, Gamma Exposure for High dose rate, or Condition D for low dose rate				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 70 V dc	I_{CBO1}		20	µA dc
Cutoff current, emitter to base	3061	Bias condition D; V_{EB} = 6 V dc	I_{EBO1}		20	µA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	60		V dc

See footnotes at end of table.

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* TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 45 V dc	I_{CBO2}		4	nA dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 5 V dc	I_{CEO1}		4	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO2}		4	nA dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 µA dc	[h_{FE1}] 5/			
2N2919, 2N2919L, 2N2919U				[30]	240	
2N2920, 2N2920L, 2N2920U				[87.5]	600	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 100 µA dc	[h_{FE2}] 5/			
2N2919, 2N2919L, 2N2919U				[50]	325	
2N2920, 2N2920L, 2N2920U				[117.5]	800	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 mA dc;	[h_{FE3}] 5/			
2N2919, 2N2919L, 2N2919U				[75]	600	
2N2920, 2N2920L, 2N2920U				[150]	1,000	
Base-emitter saturation voltage	3066	Test condition A; I_C = 1.0 mA dc; I_B = 100 µA dc;	$V_{BE(sat)1}$	0.50	1.15	V dc
Collector-emitter saturation voltage	3071	I_C = 1.0 mA dc; I_B = 100 µA dc;	$V_{CE(sat)1}$		0.35	V dc
Forward-current transfer ratio (gain ratio)	3076	V_{CE} = 5 V dc, I_C = 100 µA dc (see 4.5.5)	$\frac{h_{FE2-1}}{h_{FE2-2}}$	0.8	1.25	

See footnotes at end of table.

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* TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Absolute value of base-emitter-voltage differential	3066	Test condition B; V_{CE} = 5 V dc, I_C = 10 µA dc (see 4.5.6)	$\lvert V_{BE1} - V_{BE2} \rvert_1$		10	mV dc
Absolute value of base-emitter-voltage differential	3066	Test condition B; V_{CE} = 5 V dc, I_C = 100 µA dc (see 4.5.6)	$\lvert V_{BE1} - V_{BE2} \rvert_2$		6	mV dc
Absolute value of base-emitter-voltage differential	3066	Test condition B; V_{CE} = 5 V dc, I_C = 1 mA dc (see 4.5.6)	$\lvert V_{BE1} - V_{BE2} \rvert_3$		10	mV dc

1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to the corresponding L and U suffix versions unless otherwise noted.
4/ See 6.2.e herein.
5/ See method 1019 of MIL-STD-750 for how to determine [h_{FR}] by first calculating the delta ($1/h_{FE}$) from the pre- and Post-radiation h_{FE}. Notice the [h_{FE}] is not the same as h_{FE} and cannot be measured directly. The [h_{FE}] value can never exceed the pre-radiation minimum h_{FE} that it is based upon.

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TABLE III. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 and 4.5.9 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc, 6,000 cycles, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.	
Electrical measurements		See table I, subgroup 2 and 4.5.9 herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See 4.3.1	
Subgroup 5			
Not applicable			
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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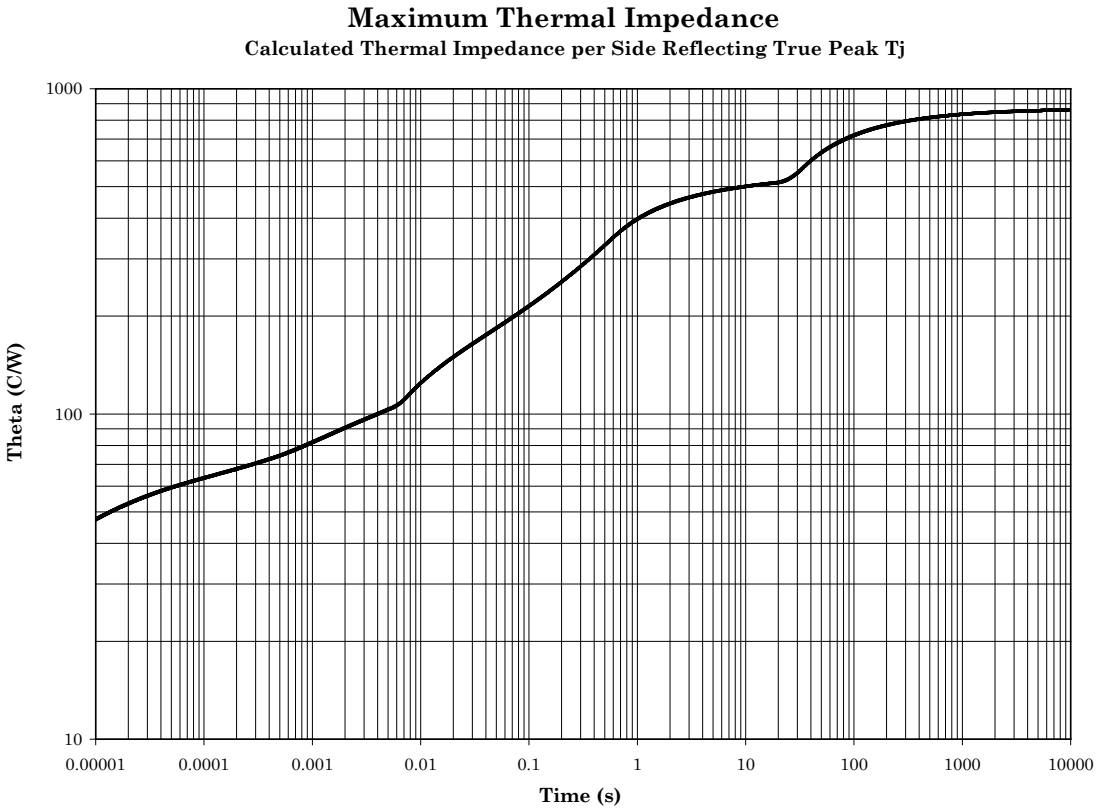


FIGURE 6. For each side: Thermal impedance = 875°C/W, Pt = 200 mW.

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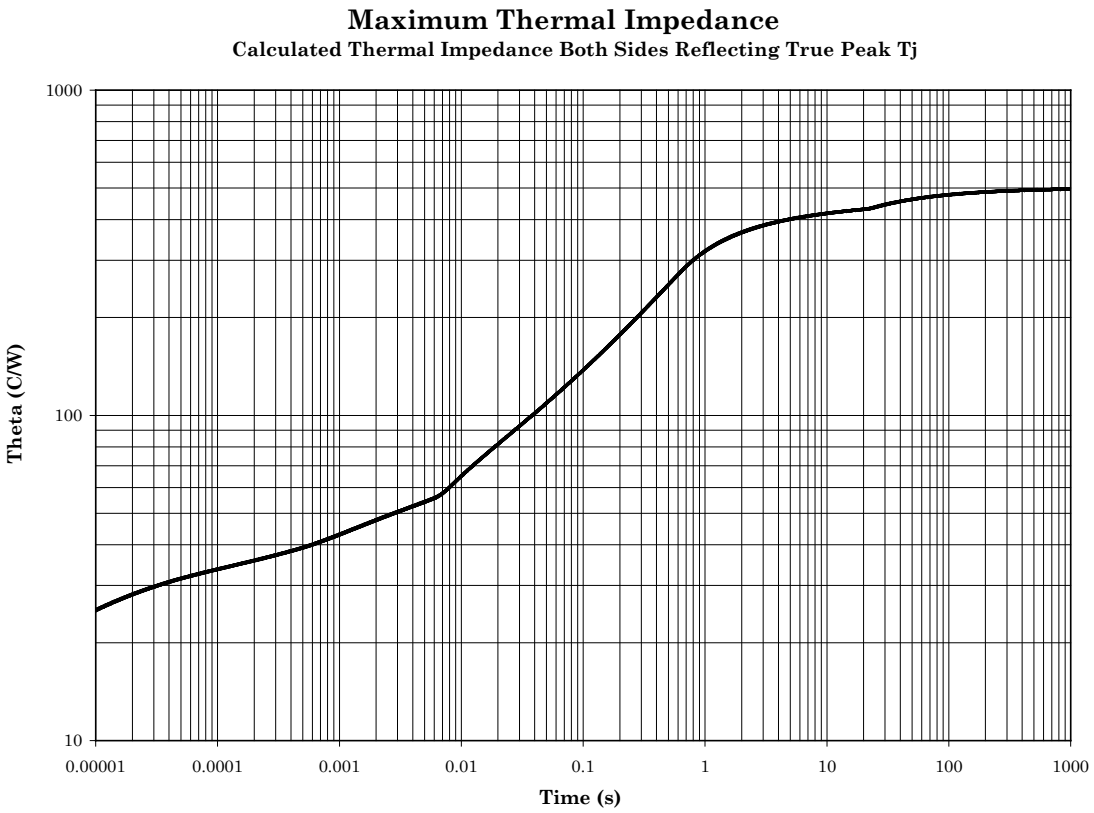


FIGURE 7. Both sides: Thermal impedance = 500°C/W, Pt = 350 mW.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

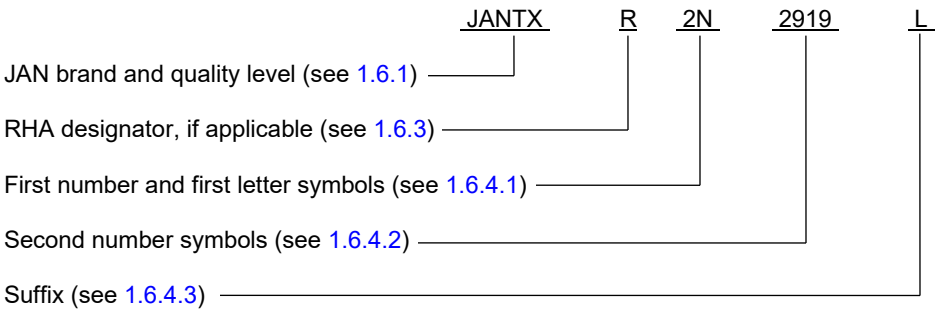
d. The complete PIN, see 1.6 and 6.5.

e. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it should be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example.

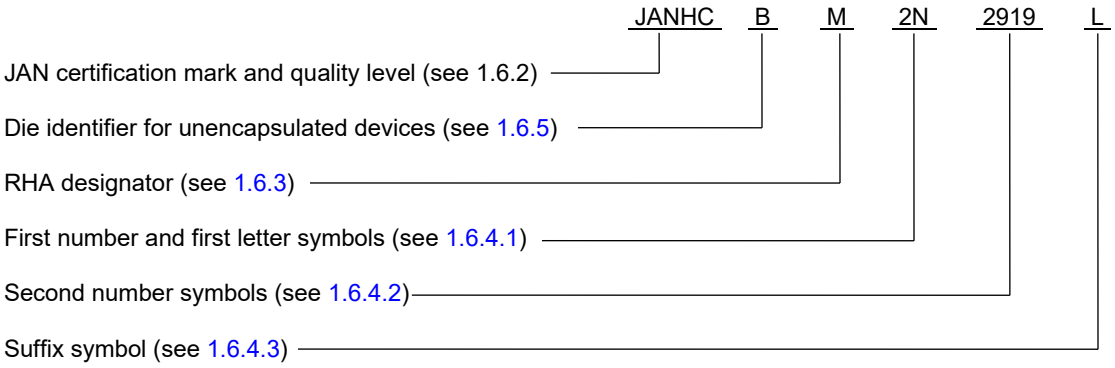
6.4.1 Encapsulated devices. The PINs for encapsulated devices are construction using the following form.



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6.4.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.

*



6.5 List of PINs.

6.5.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level (1)	PINs for devices of the "S" quality level (1)
JAN2N2919	JANTX2N2919	JANTXV#2N2919	JANS#2N2919
JAN2N2919L	JANTX2N2919L	JANTXV#2N2919L	JANS#2N2919L
JAN2N2919U	JANTX2N2919U	JANTXV#2N2919U	JANS#2N2919U
JAN2N2920	JANTX2N2920	JANTXV#2N2920	JANS#2N2920
JAN2N2920L	JANTX2N2920L	JANTXV#2N2920L	JANS#2N2920L
JAN2N2920U	JANTX2N2920U	JANTXV#2N2920U	JANS#2N2920U

* (1) The number sign (#) represent one of eleven RHA designators available (E, K, U, M, D, P, L, R, F, G, or H) if desired. Remove for no RHA.

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* 6.5.2 List of PINs for unencapsulated devices. The following is a list of possible PINs available on this specification sheet along with the supplier. The qualified JANHC suppliers with the applicable letter version (example JANHC#A2N2919) will be identified on the QML-19500.

JANHC and JANKC ordering information				
PIN	Manufacturer			
	43611	34156	9N185	52GC4
2N2919	JANHCA#2N2919, JANKCA#2N2919	JANHCB#2N2919, JANKCB#2N2919	JANHCC#2N2919, JANKCC#2N2919	JANHCD#2N2919, JANKCD#2N2919
2N2920	JANHCA#2N2920, JANKCA#2N2920	JANHCB#2N2920, JANKCB#2N2920	JANHCC#2N2920, JANKCC#2N2920	JANHCD#2N2920, JANKCD#2N2920

* (1) The number sign (#) represent one of eleven RHA designators available (E, K, U, M, D, P, L, R, F, G, or H) if desired. Remove for no RHA.

* 6.6 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2023-067)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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